

Developing Collaborative Technologies for Interdisciplinary Care Teams: What Does This Mean in the Case of Medication Reconciliation?

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Abstract

During care transitions within the hospital, clinical teams align new prescriptions with previous treatment plans through medical reconciliation. Although intended to be a safe process, almost one in three patients is subject to at least one medication error at admission or discharge. Current hospital systems do not adequately support collaborative work and instead of integrating into clinical workflows, they oftentimes require additional, potentially disruptive actions. The goal of my thesis is to explore how to build collaborative technologies for care transitions, illustrated by the example of medication management. I have conducted extensive field work in three different hospital environments, and identified how physicians repurpose distributed systems for informal communication or dealing with information overload. Building on these insights, my next step involves conducting participatory design workshops with care providers and evaluate how to successfully design tools that take into account this appropriation. This thesis aims to contribute to the understanding how to design collaborative technologies for safety critical environments.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

hospital studies, collaboration, clinical interfaces

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1 Introduction

Hospital physicians must collaborate within multidisciplinary care teams to ensure safe transitions during admission, ward transfers,

and discharge. Medication reconciliation defines the formal process for aligning a patient's medication orders to avoid omissions, duplications, dosing errors, or drug interaction [5].

Nevertheless, almost one out of three patients is subject to one or more medication errors at hospital admission or discharge [11]. Of these errors, 26 % are considered *serious or very serious* and the risk for an incident increases with the number of medications. The use of electronic health records (EHR) presents a complex dilemma: they reduce certain errors such as wrong orders but simultaneously introduce new error due to system configuration problems [19]. Hospital technologies often suffer from exceeding complexity, low usability, and a lack of training and support for clinical staff [16]. Furthermore, they usually address single users and do not adequately account for the interdisciplinary and socio-technical nature of hospital work [2]. In my thesis, I explore how to build collaborative technologies for interdisciplinary care teams, illustrated by the use case of medication reconciliation. Based on empirical insights from the field and user perceptions, I will conduct participatory design workshops and involve clinicians to discuss and re-imagine digital tools for safer medication management. Hereby, I aim to apply aspects of “*generative theories of interaction*” [6] through the lens of Common Information Spaces as introduced by Schmidt and Bannon [22] and adapted to the hospital ward by Bossen [8].

2 Related Work

2.1 Hospital Technologies for Effective Medication Reconciliation

Available information is distributed over fragmented sources and in order to ‘*achieve an overview*’ physicians need to manage and combine information from different digital and analogue artifacts [9]. The design of electronic health records is static, inadequately addressing the varied information needs associated with a wide range of tasks [14]. Several application for medication reconciliation have been proposed in previous work [13], but they still lack interoperability, “user-friendly” information and the integration with the ordering process.

Clinical Decision Support (CDS) Tools potentially offer more support for medication reconciliation, but usually target only one user [17] and do not account for the collaborative aspects of hospital work [10, 17]. Managing medications for patients with multiple health conditions requires the perspective of different specialties who can propose different care alternatives[12].

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2.2 Common Information Spaces Across Distributed Clinical Teams

Schmidt & Bannon have introduced common information spaces through the “artefacts that are accessible to a cooperative ensemble as well as the meaning attributed to these artefacts by the actors.” [22].

They are particularly relevant in collaborative settings where multiple stakeholders need to access, interpret, and contribute to shared data or knowledge. This concept has been widely picked up and refined for the hospital setting [4, 8, 29]. In a hospital ward, common information spaces are dynamically constructed through digital and analogue artifacts [3]. The mediating qualities of common information spaces are an interesting and crucial characteristic to support care transitions [6] and further research is required to examine how to translate this to distributed care teams.

2.3 Participatory Design With Clinicians for Better Workflow Integration

How to include users into the design process is not a new question in the HCI-community, but persists a contemporary debate in the field of healthcare [1]. An ongoing challenge remains how to engage already overcharged staff in research studies [7, 15, 23] and evaluate solutions effectively [27]. Previous studies that conducted co-design workshops with clinicians have highlighted the need for flexibility and adaptability to address challenges such as stakeholder engagement and power dynamics [10, 18, 24, 25]. Key learnings from these studies underscore that engaging a broad range of perspectives ensures that potential barriers to adoption are identified early, facilitating smoother integration into existing systems. Especially in the case of collaborative technologies, this approach can enhance the perceived relevance and usability and promote a sense of ownership among users [10].

3 Research Goals

The objective of my thesis is to understand how to build collaborative technologies for care transitions illustrated by the example of medication management. Current information systems in hospitals are distributed, usually target a single user and often add further potentially disruptive work steps for care providers [26]. With this in mind, I would like to explore from an HCI-perspective, how collaborative systems can support coordination of care transitions in interdisciplinary teams. This work is guided by the following three goals:

- (1) **Understand:** What are clinicians’ strategies when using or appropriating available technologies or artifacts, to mitigate medication errors during care transitions? Which information do providers need for effective medication reconciliation during such transitions?
- (2) **Participatory Design:** Based on this understanding, how can we design collaborative tools that support common understanding of patient data and effective medication reconciliation?
- (3) **Evaluate:** Moving beyond the lab, how to successfully implement such collaborative medication tools in hospital environments?

4 Current work and outcomes

I have conducted extensive field work in three different hospitals to understand care processes and boundaries of technology to support them. I was able to shadow clinicians in a German, U.S., and a French hospital, gathering insights across different healthcare systems.

How to design for effective medication management in interdisciplinary care teams persisted a key question in these environments, regardless of implemented IT-systems. The following section gives a summary of my approach and main takeaways that motivate the following steps of the thesis.

4.1 Research Approach and Methodology

The observations in the German hospital were part of an observation study that aimed to understand how physicians retrieve and integrate heterogeneous data into care. A University Internal Review Board (IRB) approved the study and some insights were published in [28]. The observations in the U.S. hospital were conducted as part of a quality improvement initiative and the site visits at the French hospital were exploratory observations. For IRB reasons, I cannot report data on these observations. However, these observations guided my research focus and helped me develop a better understanding for the hospital environment and the different roles in medical teams.

4.1.1 Hospital environments and participants. The German study took place at a regional 360 bed acute care hospital. I observed members of the internal medicine department, who iterate over multiple wards (cardiology and pulmonology, gastroenterology, emergency care and intensive care unit). Throughout four months, I followed the shifts of two senior physicians and nine residents and systematically documented their interactions with information systems.

The U.S. observations took place in a large urban specialized academic hospital with a total of 2000 inpatient bed across multiple clinics. The observations in France took place in two cancer centers in a urban area. For these comparative observations, I followed total of seven different physicians from internal medicine, anesthesiology and surgery.

4.1.2 Data Collection Structure. We developed a data collection tool using macros in the Apple Shortcuts application on an iPhone 12 Pro, which we call SNAPCuts. SNAPCuts structures data collection into adaptable pre-defined templates with the option to capture meta information such as time automatically. The observation templates were designed to capture physician’s interactions with technology in a map between available data sources (e.g., hospital information system, text editor, letter, personal notes and data types (e.g., medication, laboratory results, next procedures).

The studies focused on how physicians retrieve patient data during their shifts and distinguish among different data sources and types. We wanted to create a map of these sources and types, and included a temporal component to capture how different rhythms during the day shape the interactions [21]. We adapted the templates to each hospital setting and during the study, I followed one physicians throughout their shift and documented observations systematically. This approach supported my data collection for over

144 hours in the German hospital, about 30 hours in the United States and 15 hours in France. For future work, I am interested to explore how *SnapCuts* can be extended to other use cases, and I am currently collaborating with researchers from NYU who introduce the flexible templates of *SnapCuts* to personal tracking for patients with multiple sclerosis.

4.2 Findings and Important Takeaways

The extensive field work over the last 1.5 years has helped me to develop a better understanding of the work in the hospital environment and in the following, I will highlight my key takeaways that have shaped the focus of my thesis.

Physicians repurpose distributed systems for functions beyond their original design

Physicians in the hospital ward have access to a variety of information systems to access relevant data about a patient. I observed different strategies, how to structure and organize this information. Most physicians keep a printed patient list with the most important results and tasks that they carry with them. In accordance with previous research [3], the location of physical copies indicate tasks or patient information. The medical discharge letter is not only treated as a final summary, but rather like a working document, that summarizes the patient stay and can include implicit messages on further treatments [28].

Physicians fall back to interpersonal communication, but it's not that easy

Most of relevant information in the German observation study was exchanged through interpersonal communication and I was surprised how much clinicians relied on their memory to distinguish patient cases. Each morning started with an analogue handover from the night shift, in which the responsible resident presented new patients and highlighted if critical medication like antibiotics were administered upon admission. The other physicians noted these changes on personal paper notes and mainly used these paper notes to keep the most important information.

The process to find relevant information from electronic health records oftentimes resembled “looking for a needle in a haystack” [28] and physicians oftentimes called or referred to their colleagues to ask for an information that they were looking for.

Medication Information Needs to Be Manually Copied Across Distributed Systems

When a patient is admitted or discharged to the hospital, many patients already have a medication regime in addition to newly prescribed drugs. At these transitions, healthcare providers must conduct thorough medication reconciliation to create an accurate list of a patient's current medications. Currently, this process is performed by manually copying this information across scattered systems and documents. The wrong medication or dosage is unfortunately only one click away and I observed several situations, in which hospitalized patients obtained e.g. the medication of the room neighbor, a continuous administration of an infusion that intended to be a single one, or missing prescriptions.

One particular situation in the German hospital inspired my further explorations. One morning, a resident shared with me that when he reviews patients, he begins by examining their medications and asking himself, “*Why is this patient taking this medication?*” for

each prescription. I observed, that every resident had their own approach in familiarizing themselves with the patients and the provided systems do not guide them through this task.

5 Next steps

The observations of my previous work provided me a profound understanding of daily hospital work in different healthcare systems. Led by curiosity, this research started very open-ended, but with more time and experience, the coordination of medication plans stood out as a frequent breakdown. In the next steps of my thesis, I want to concentrate on the design and use of collaborative tools during care transitions. The use case of medication reconciliation offers an interesting, yet critical setting, to explore how to include technologies for better care.

5.1 Participatory Design Workshops With Clinicians to Examine Collaborative Tools for Medication Reconciliation

During my next study, I aim to build upon the insights from my ethnographic work and focus on how to build collaborative tools for interdisciplinary care teams, illustrated by the use case of medication reconciliation. Previous observations have provided me an overview on what is happening, and in the next steps I would like to concentrate more on the *why*. Prior to the workshops, I would thus like to conduct a short interview study with clinicians to identify their personal strategies to successful reconciliation management and systematically extract barriers of current technologies to support this as a collaborative process. These focused interviews will provide me a new angle with which I can revisit my observation data and design a protocol for participatory design workshops. As clinicians only have limited time and resources to participate in research studies, the priority of the workshop design is an easy access and participation. I consider to explore how *SNAPCUTS* can support me in this process and pre-structure resources in advance. Ideally, the outputs of these workshops will be video prototypes [20] or probes that show potential interaction pathways.

5.2 Moving Beyond the Lab: Evaluate Collaborative Medication Tools for the Hospital

Evaluating collaborative medication tools in hospitals from a Human-Computer Interaction (HCI) perspective involves understanding how these tools facilitate user engagement, improve medication adherence, and enhance communication among healthcare providers and patients. Through the participatory design workshops with users I aim for a user-centered design approach and in the last part of my thesis, I will investigate potential evaluation methods. The lens of generative theory of interaction [6] will serve for a design that is grounded in theory and empirical data. By establishing clear criteria for evaluating these tools—such as patient orientation, functionality, security, and user satisfaction—researchers can better assess their impact on medication management outcomes. Ultimately, the goal is to create evidence-based guidelines that inform the design and implementation of collaborative medication tools in hospital settings, promoting safer and more effective patient care.

6 Dissertation status and long term goals

I am currently a second-year PhD. candidate at Paris-Saclay University supervised by Wendy E. Mackay and Ignacio Avellino and I aim to defend my thesis by Summer 2026. I have never attended a doctoral consortium at previous SIGCHI conferences and I appreciate this initiative for the chance to learn about diverse PhD research paths. I am especially keen to discuss methodological approaches for HCI studies in safety-critical environments, such as the hospital.

My long-term goal is to work in research at the intersection of HCI and Healthcare and investigate solutions for clinical interfaces that support physicians to obtain the right information at the right time. Through my research and academic service, I hope to contribute to a curious and interdisciplinary HCI community and work towards a permanent academic position in the future.

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